Journal of Physical Education and Sport ® (JPES), Vol. 25 (issue 9), Art 213, pp. 1919-1930, September 2025
online ISSN: 2247 - 806X; p-ISSN: 2247 – 8051; ISSN - L = 2247 - 8051 © JPES

Original Article

Comparison of resistance training frequencies (2, 3, and 4 days/week) on strength, body composition, and anthropometric measures in untrained young men: An equal-set, intensity-matched randomized controlled trial

TAVARINTORN RUKBUMRUNG[1], MANINTHORN RUGBUMRUNG[2*], ADISON THURAYOT[3], SITTIWIT IMPANYA[4], PICHET CHAILERT[5], PATCHARIN TANGCHAISURIYA[6]
[1]Department of Physical Education, Faculty of Education, Prince of Songkhla University, Pattani Campus, THAILAND
[2,3,4,5,6,]Sport and Exercise Science Program, School of Science, University of Phayao, THAILAND

Published online: September 30, 2025
Accepted for publication: September 15, 2025
DOI:10.7752/jpes.2025.09213

Abstract
This randomized controlled trial investigated the impact of different resistance training (RT) frequencies (2, 3, or 4 days/week) on muscular strength, body composition, and anthropometric measures in untrained young men. Training volume and intensity were matched across groups. Over an 8-week period, 41 previously untrained male university students and staff (aged 18–35 years; BMI 18.5–24.9 kg/m^2) participated. Participants were randomly assigned to one of three training groups (2, 3, or 4 sessions per week) or to a control group. All training groups completed 12 sets per exercise per week, with intensity individualized based on one-repetition maximum (1RM) using a Daily Undulating Periodization (DUP) model. Key outcome measures included muscular strength (1RM in the barbell chest press, barbell row, Smith machine squat, and hip thrust), body composition (body weight (BW), fat-free mass (FFM), and body fat percentage (%Fat) assessed via bioelectrical impedance analysis (BIA)), and anthropometric measures (chest, upper-arm, and thigh circumferences). All variables were assessed before and after the training period. Data analysis involved paired-sample t-tests to evaluate within-group changes and an analysis of covariance (ANCOVA) to compare between-group differences, with baseline values entered as covariates. Results indicated that all training groups demonstrated statistically significant improvements in muscular strength (1RM across all exercises), body circumferences, and FFM following the intervention ($p < 0.05$ for all, with large effect sizes). Post-hoc analyses showed that all three training groups achieved significantly greater improvements in the measured variables compared with the non-training control group ($p < 0.05$ for all). However, no significant differences were observed among the 2-, 3-, and 4-day training groups for any outcome (muscular strength, body circumferences, or FFM). These findings suggest that when total weekly volume and relative intensity are matched, RT performed 2, 3, or 4 days/week produces comparable gains in strength, body composition, and anthropometric measures in previously untrained young men.
Keywords: weight training, hypertrophy, physical fitness

Introduction

Resistance training (RT) is a key form of anaerobic exercise that effectively improves physical fitness, particularly muscular strength, power, and size, while also improving body composition (ACSM, 2018; Schoenfeld, 2010). It is widely practiced and extensively studied owing to its broad benefits, which include improving appearance, balancing body proportions, reducing fat mass, increasing bone and muscle mass, increasing resting metabolic rate, and aiding in weight management. RT has also been shown to improve physical performance in both athletes and non-athletes (Hong & Kim, 2018). Studies suggest that performing RT more than once per week can increase both muscular strength and size (Schoenfeld et al., 2016). The American College of Sports Medicine (ACSM, 2018) recommends that healthy adults perform RT at least 2–3 days/week to gain physiological benefits. However, participation rates remain low. Even in countries with strong exercise promotion, such as the United States, fewer than one-third of adults engage in RT on a regular basis (Paluch et al., 2024).

Effective muscle training initially increases strength primarily through neural adaptations. With continued training, muscle mass gradually increases, becoming a major contributor to further strength gains. Among highly trained or elite athletes, neural adaptations eventually plateau, resulting in minimal additional gains unless new training stimuli are introduced (Hughes et al., 2018). In untrained individuals, however, the progression of these adaptations is less well understood. Studies suggest that measurable muscle hypertrophy typically emerges after at least four weeks of consistent training (Rhea et al., 2002).

The effectiveness of RT is influenced by several factors, including training volume (the number of sets, repetitions, and load) and training frequency (how often each muscle group is trained per week) (Wernbom et al.,

Corresponding Author: MANINTHORN RUGBUMRUNG, E-mail: physicalfitnessthailand@gmail.com

2007; [2]Schoenfeld et al., 2019). Although training volume is often considered the primary determinant of adaptations, training frequency is also important and has received considerable research attention because it directly affects the time commitment required—an essential factor for maintaining consistency in both novice and experienced trainees.

Research on the effects of RT on muscle strength and size is extensive but remains inconclusive. Many earlier studies examined scenarios in which total training volume was carefully matched (Moesgaard et al., 2022). In these volume-matched studies, some reported that higher training frequencies (e.g., 3–4 sessions per week) produced greater improvements (Grgic, 2018; McLeod et al., 2024; Schoenfeld et al., 2016), whereas others found no significant differences based on frequency (Kessinger et al., 2020; Cuthbert et al., 2021; Hamarsland et al., 2022; Neves et al., 2022). Similar inconsistencies are observed in studies involving untrained participants: some suggest that higher frequency yields superior results (Cigerci & Genc, 2020), while others report no meaningful effect when volume is controlled (Hansen et al., 2017; [1]Schoenfeld, 2019; Zaroni, 2019). These conflicting findings indicate that study design may influence outcomes because volume-matched protocols could obscure the true effects of training frequency.

Beyond controlling for training volume, researchers can also standardize training frequency and intensity using an equal-set, intensity-matched design. In this approach, all groups perform the same total number of sets at the same relative intensity—often using a percentage of one-repetition maximum (1RM). This allows comparison of protocols such as fewer sets at higher frequency (e.g., 3 sets per session, 4 sessions per week) versus more sets at lower frequency (e.g., 6 sets per session, 2 sessions per week), with intensity held constant (e.g., 70% 1RM) (Arazi et al., 2021). The main advantage of this approach is that it enables researchers to assess how varying set numbers and training frequencies affect physical gains, thereby identifying the most effective strategy for beginners or individuals new to resistance training.

Therefore, this study investigates the effects of different RT frequencies (2, 3, or 4 days/week) using an equal-set, intensity-matched design in untrained young male adults. Specifically, it assesses how training frequency influences muscular strength (1RM in the barbell chest press, barbell row, Smith machine squat, and hip thrust), body composition (body weight (BW), body fat percentage (%Fat), and fat-free mass (FFM)), and anthropometric measures (chest, upper arms, and thighs). This design addresses a gap in the existing literature, which provides limited data on the effects of training frequency in this population.

Materials and methods

Study design

This study used an 8-week randomized controlled trial (RCT) design and was approved by the University of Phayao Human Experimental Ethics Committee (Project No. UP-HEC 1.3/023/64). Sample size calculations were performed using G*Power 3.1 Assuming a medium effect size (ES) = 0.6 for muscular strength estimate based on the reported an effect size greater than 0.8 for strength improvements when training occurred at least twice per week (Cigerci et al., 2018) with 80% power and a significance level of 0.05, the analysis indicated that 35 participants would be required to detect a statistically significant effect. To account for potential dropouts, the sample size was increased to 48 participants, with 12 individuals assigned to each group. Participants were randomly assigned to one of four groups using a simple randomization procedure, such as drawing sealed opaque envelopes. The groups were: Experimental Group 1, performing RT two days/week; Experimental Group 2, training three days/week; Experimental Group 3, training four days/week; and a Control Group, which received no specific intervention.

Participants

Participants were recruited from male students and staff volunteers at the University of Phayao. Inclusion criteria were: age 18–35 years, selected to minimize the influence of hormonal changes on muscle hypertrophy, which typically decline after age 35 (Tan et al., 2023); a body mass index (BMI) of 18.5–24.9 kg/m^2, chosen to ensure homogeneous hypertrophic responses and represent a healthy weight range (Tan et al., 2023); and no RT experience in the previous six months (Lasevicius et al., 2022). Exclusion criteria included any health conditions or illnesses that could compromise safe participation or hinder compliance with the study protocol; a history of heart disease or severe hypertension (blood pressure ≥180/100 mmHg); and current use of medications or substances known to influence muscle growth or metabolism. All volunteers completed a comprehensive health history questionnaire to screen for pre-existing conditions. Eligible participants were required to provide written informed consent prior to voluntarily joining the study.

Intervention

Participants in the intervention groups completed an 8-week (Cigerci et al., 2018; Chaves et al., 2020) supervised RT program at the fitness center of the Department of Exercise Science and Sport, University of Phayao. Training sessions were performed in a temperature-controlled environment, maintained between 25°C and 27°C.

The experimental period lasted 11 weeks: Week 1 served as the pre-test (baseline); Week 2 involved familiarization with the training program; Weeks 3–10 comprised the 8-week intervention; and Week 11 was the post-test period. Each 4-week block was treated as a mesocycle. The 1RM test was performed during the pre-intervention period and at the start of each new mesocycle to individualize training loads. This program was designed to ensure that all groups received an equivalent training stimulus, with equal intensity (based on a percentage of each participant's 1RM) and an equal number of sets (equal sets) for each exercise per week. This approach differs from protocols that control for "total volume load," which equalize the overall weight lifted across groups. Instead, it accounts for individual differences in strength and prior training experience, which is critical for designing effective, practically implementable programs that support long-term adherence, particularly in untrained populations. RT program design involves key variables—intensity, volume, and progression—all of which should be tailored to each individual (Schoenfeld, 2016). In this study, intensity was held constant throughout each mesocycle, with exercises including barbell chest press (pectoralis), barbell row (latissimus dorsi), bicep curl (biceps), cable push down (triceps), Smith machine squat (quadriceps), and hip thrust (gluteus). The exercise intervention involved 12 sets per exercise each week, totaling 96 sets per exercise across 8 weeks for all experimental groups.

Each experimental group followed daily undulating periodization (DUP), which involves daily variations in volume and intensity (Rhea et al., 2002) for all assigned exercises. All weight training sessions were separated by at least 48 h of recovery, except in the 4-day training group, where one session had a recovery interval shorter than 48 h but not less than 24 h. ***Experimental Group* 1** (2 days/week): Participants followed an Intra-Session Undulating Periodization (IUP) model, in which intensity and volume varied within the same training session. Each training day consisted of six sets per exercise, totaling 12 sets per exercise per week. On Training Day 1, sets were structured as 8–10 RM (Sets 1–2), 6–8 RM (Sets 3–4), and 4–6 RM (Sets 5–6). On Training Day 2, the sequence shifted to 10–12 RM (Sets 1–2), 8–10 RM (Sets 3–4), and 6–8 RM (Sets 5–6). ***Experimental Group 2*** (3 days/week): This group followed a DUP model, performing four sets per day for a total of 12 sets per exercise per week. Each training day emphasized a specific RM range for all sets: 8–10 RM on Training Day 1, 4–6 RM on Training Day 2, and 10–12 RM on Training Day 3. ***Experimental Group 3*** (4 days/week): This group also followed a DUP model, performing three sets per day for a total of 12 sets per exercise per week. The program rotated RM ranges across the four sessions: 8–10 RM on Training Day 1, 10–12 RM on Training Day 2, 4–6 RM on Training Day 3, and 8–10 RM on Training Day 4. These RM ranges corresponded to varying percentages of 1RM daily, ensuring equal intensity across groups. Each set was performed to momentary concentric muscular failure. A 90-s rest interval was given between sets and a 180-s rest interval between exercises. The external load (kg) was adjusted as needed for each set to ensure participants reached failure within the target repetition range. Before the first set of each exercise, participants performed a warm-up of 2 sets of 10 repetitions at 50% of the working load (Corrêa, 2022).

Participants were instructed to maintain their usual daily routines, refrain from taking supplements during the study, and follow their customary dietary intake.

Control Group: Participants in the control group maintained their usual daily activities without engaging in any structured exercise program.

Testing

All testing was performed during the pre-intervention period (Week 1). Prior to testing, participants were instructed to maintain their usual daily routines and to refrain from strenuous physical activity, alcohol consumption, and smoking for at least 48 h (van den Tillaar & Ball, 2020). They were also asked to avoid food and beverages for at least 2 h before performing the physical fitness tests.

1) 1RM Test: The 1RM test was performed over two separate days to minimize participant fatigue and ensure measurement reliability. One week prior to the initial test, familiarization sessions were held, especially for participants with no prior RT experience. These sessions allowed participants to practice the test exercises, refine their technique, and learn to exert maximal force to ensure accurate 1RM measurements. Participants were also instructed to avoid intense exercise for at least 48 h before each testing day. On Day 1, participants were tested on the barbell chest press, Smith machine squat, and cable push-down, while Day 2 assessments included the barbell row, bicep curl, and hip thrust. Each testing day began with a 15-min warm-up, consisting of 5 min of treadmill jogging followed by 10 min of dynamic stretching for both the upper and lower body. This was followed by specific warm-up sets: one set of 8–10 repetitions at 50% of the estimated 1RM and one set of 2–3 repetitions at 60–80% of the estimated 1RM. For the 1RM determination, participants performed successive attempts with progressively heavier loads, resting 3–5 min between attempts, all 1RM determinations were made within five attempts (Mao, 2023). To further ensure the robustness of the data, each exercise was tested twice, with the second measurement performed 48 h after the initial test for that specific exercise, and the final 1RM value was determined by averaging both test results (Grgic et al., 2020; Johnsen & van den Tillaar, 2021).

2) Body Composition: BW, %Fat, and FFM were assessed using bioelectrical impedance analysis (BIA) with the ACCUNIQ BC380 device (Yang et al., 2018). Participants wore standardized light clothing without accessories, jewelry, shoes, or socks. Measurements were consistently taken in the morning, prior to consuming food or beverages and after voiding.

3) Anthropometric Measures (Resting Body Circumference): Body circumferences were measured with a metric tape accurate to 1 mm (Staniszewski et al., 2020). To minimize measurement error (kept below a 2%

margin), all assessments were performed by the same trained assessor (Wang et al., 2000; Staniszewski et al., 2020). The following sites were measured:

- Upper arms (left and right): Measured at the mid-biceps, halfway between the acromion and olecranon processes.
- Chest: Measured at maximal exhalation.
- Thighs (left and right): Measured at mid-thigh, halfway between the inguinal crease and the superior patella.

Ethics Approval

This study was approved by the University of Phayao Human Experimental Ethics Committee, Thailand, in accordance with the Declaration of Helsinki, Project No. UP-HEC 1.3/023/64.

Statistical Analysis

All statistical analyses were performed using SPSS (Version 26.0). Data normality was assessed using the Kolmogorov–Smirnov test. Descriptive statistics for participant characteristics (age, height, and weight) are reported as mean ± standard deviation (SD). To examine baseline differences between groups for all physical performance measures including body composition, body circumferences, and 1RM in the barbell chest press, barbell row, Smith machine squat, and hip thrust one-way analysis of variance (ANOVA) was performed. The primary analysis to compare the effects of the interventions between groups after 8 weeks was performed using analysis of covariance (ANCOVA). This approach allowed post-intervention values of the dependent variables including 1RM in the barbell chest press, barbell row, Smith machine squat, and hip thrust, as well as body composition and body circumferences to be compared while adjusting for baseline values as covariates. When ANCOVA revealed significant main effects, post-hoc comparisons were performed using the Bonferroni correction to identify specific between-group differences.

Additionally, to quantify the magnitude of change in each group, percentage changes from baseline to post-intervention were calculated using the formula: ((Mean after eight weeks of intervention − Mean at baseline) / Mean at baseline) × 100. The practical significance of within-group changes was quantified using Cohen's *dz*, calculated as D / SD (D), where D is the mean of the difference scores and SD (D) is the standard deviation of the differences. Cohen's *dz* values were interpreted according to Cohen's (1988) guidelines: 0.2 represents a small effect, 0.5 a medium effect, and 0.8 a large effect.

Results

Participants and Intervention

The study began with 48 volunteer participants in the first week, who were randomly assigned to four experimental groups and one control group. Each experimental group completed 8 weeks of weight training according to their prescribed protocol, while the control group continued their usual daily routines without any exercise intervention. At the end of the 8-week period, 41 participants remained, including 31 from the experimental group and 10 from the control group. Data on age, weight, and height for these 41 participants, collected during testing in the first week, are presented in Table 1.

Table 1. Participant characteristics at baseline

Variable	2 days weight (n = 10)		3 days weight (n = 10)		4 days weight (n = 11)		Control (n = 10)	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
Age (years)	20	± .00	21.60	± 1.27	21.07	± .47	20.50	± .52
Height (cm)	173.80[1]	± 4.47[1]	172.70[1]	± 5.62[1]	171.09[1]	± 5.54	170.40[1]	± 4.03
Weight (kg)	65.90[1]	± 6.27	66.61[1]	± 8.71	65.55[1]	± 7.25	65.66[1]	± 8.81

[1] = no statistically significant difference between groups as determined by ANOVA

Impact of Weight Training on Body Composition

Weight training produced direct effects on BW and FFM, while no significant changes were observed in %Fat. Results from paired-sample t-tests demonstrated a significant increase in BW and FFM across all experimental groups after the 8-week intervention, as shown in Table 2. **BW:** *2 Days/week Training Group*; baseline (65.90 ± 6.27) to post-test (66.68 ± 5.85), $t(9) = 2.607$, $p < .05$, $d = .82$); *3 Days/week Training Group*; baseline (66.61± 8.26) to post-test (67.36 ± 8.97), $t(9) = 2.583$, $p < .05$, $d = .81$; *4 Days/week Training Group*; baseline (65.55 ± 7.25) to post-test (66.28 ± 6.74), $t(10) = 3.172$, $p < .05$, $d = .95$. **FFM:** *2 Days/week Training Group*; baseline (53.36 ± 4.91) to post-test (54.14 ± 5.08), $t(9) = 4.949$, $p < .05$, $d = 1.56$); *3 Days/week Training Group*; baseline (53.54 ± 5.37) to post-test (54.30 ± 6.13), $t(9) = 2.305$, $p < .05$, $d = .73$; *4 Days/week Training Group*; baseline (52.30 ± 6.07) to post-test (53.01± 5.98), $t(10) = 3.479$, $p < .05$, $d = .1.05$. No significant differences were observed in the control group for FFM.

ANCOVA revealed no significant differences in BW and %Fat between the experimental and control groups. However, after adjusting for baseline values of FFM, ANCOVA indicated a significant between-group effect ($F(3,36) = 2.998$, $p < .05$, $\eta^2 p = .200$). Post-hoc comparisons showed that, after the 8-week intervention, FFM was significantly higher in all training groups compared with the control group (all $p < .05$). No significant differences in FFM were observed among the training groups themselves (Table 3).

Table 2. Within-group comparisons of body composition, body circumferences, and muscle strength (1 RM)

Variable	**2 days/week**			**3 days/ week**			**4 days/ week**			**Control**		
	BL	**8 weeks**	**% Change (*d*)**	**BL**	**8 weeks**	**% Change (*d*)**	**BL**	**8 weeks**	**% Change (*d*)**	**BL**	**8 weeks**	**% Change (*d*)**
BC												
Weight (kg)	65.90 ± 6.27	66.68 ± 5.85	1.18* *(.82)*	66.61 ± 8.26	67.36 ± 8.97	1.13* *(.81)*	65.55 ± 7.25	66.28 ± 6.74	1.11* *(.95)*	65.66 ± 8.81	65.99 ± 8.70	.50 *(.29)*
FFM (kg)	53.36 ± 4.91	54.14 ± 5.08	1.46* (*1.56)*	53.54 ± 5.37	54.30 ± 6.13	1.42* *(.73)*	52.30 ± 6.07	53.01 ± 5.98	1.36* *(1.05)*	53.14 ± 5.50	53.11 ± 5.51	−.06 *(.23)*
% Fat	13.47 ± 2.93	13.43 ± 3.21	−.29 *(.04)*	15.80 ± 3.34	15.76 ± 3.39	−.25 *(.05)*	13.94 ± 5.54	13.97 ± 5.51	.22 *(.02)*	15.83 ± 4.84	16.05 ± 4.73	1.39 *(.30)*
Circumference (cm)												
Chest	88.28 ± 4.24	91.38 ± 3.70	3.51** *(1.73)*	92.79 ± 5.83	96.47 ± 8.65	3.97* *(.97)*	89.79 ± 4.01	93.30 ± 3.97	3.91** *(2.69)*	92.30 ± 5.52	92.95 ± 5.21	.74 *(.60)*
Upper left arm	28.19 ± 2.20	29.01 ± 2.00	2.91* *(.88)*	29.64 ± 2.54	30.73 ± 2.66	3.68** *(2.10)*	29.56 ± 3.03	30.17 ± 3.02	2.06* *(.82)*	28.96 ± 2.89	29.02 ± 2.90	.21 *(.56)*
Upper right arm	28.56 ± 2.15	29.20 ± 1.88	2.24* *(.75)*	29.72 ± 2.60	30.77 ± 2.47	3.53** *(2.40)*	29.91 ± 3.13	30.55 ± 2.84	2.14** *(1.66)*	29.11 ± 2.80	29.02 ± 2.85	−3.1 *(.66)*
Left thigh	53.56 ± 3.34	53.81 ± 3.37	.46* *(1.52)*	54.90 ± 2.92	55.34 ± 2.88	.80** *(2.48)*	54.52 ± 4.57	54.90 ± 4.54	.70** *(2.88)*	54.45 ± 6.80	54.44 ± 6.78	−.02 *(.07)*
Right thigh	52.92 ± 2.48	53.33 ± 2.57	.77** *(2.37)*	55.60 ± 3.05	55.98 ± 3.15	.68** *(2.09)*	55.05 ± 4.09	55.46 ± 4.06	.74** *(2.43)*	55.13 ± 7.04	55.12 ± 7.11	−.01 *(.07)*
Muscle strength												
1RM barbell chest press (kg)	48.97 ± 11.23	53.26 ± 15	8.76* *(.94)*	51.69 ± 10.43	56.88 ± 11.87	10.04** (*.2.42*)	50.94 ± 6.18	54.89 ± 6.18	7.75** *(2.42)*	47.91 ± 6.76	47.44 ± 7.25	−0.10 *(.10)*
1RM barbell chest press / BW	.74 ± .13	.79 ± .19	8.22* *(.81)*	.78 ± .13	.84 ± .14	7.69** *(2.93)*	.78 ± .08	.83 ± .08	6.41** *(1.87)*	.73 ± .11	.72 ± .12	−1.37 (*.19)*
1RM barbell row (kg)	44.25 ± 6.48	50.80 ± 6.98	14.80** *(2.14)*	46.74 ± 8.49	54.13 ± 9.55	15.81** *(2.18)*	45.37 ± 9.95	52.33 ± 11.21	15.34** *(2.77)*	48.73 ± 10.81	48.96 ± 10.66	0.47 (*.31)*
1RM Barbell row / BW	.67 ± .07	.7**6** ± .19	13.43** *(1.92)*	.71 ± .13	.8**1** ± .72	14.08** *(1.90)*	.69 ± .13	.80 ± .141	15.94** *(2.29)*	.74 ± .11	.74 ± .11	0 *(.03)*
1RM Smith machine squat (kg)	61.78 ± 15.41	81.17 ± 13.88	31.38* *(1.43)*	63.86 ± 13.46	83.75 ± 10.11	31.15** *(1.73)*	53.74 ± 10.00	74.38 ± 13.08	38.41** *(3.06)*	57.78 ± 10.33	58.87 ± 10.03	1.89 (.80)
1RM Smith machine squat / BW	.93 ± .18	1.23 ± .23	32.26* *(1.32)*	.95 ± .11	1.26 ± .17	32.63* *(1.42)*	.83 ± .18	1.13 ± .24	36.11** *(2.92)*	.89 ± .19	.90 ± .19	1.12 *(.61)*
1RM hip thrust (kg)	61.21 ± 6.77	68.93 ± 6.76	12.61* *(1.62)*	60.14 ± 8.60	68.38 ± 8.30	13.70** *(2.11)*	60.07 ± 8.56	68.50 ± 6.30	14.03** *(2.11)*	60.34 ± 10.39	60.36 ± 10.31	.03 *(.07)*
Hip thrust / BW	.93 ± .10	1.04 ± .09	11.83* *(1.43)*	.91 ± .10	1.02 ± .09	12.09** *(1.68)*	.92 ± .11	1.04 ± .12	13.04* *(1.47)*	.92 ± .12	.92 ± .13	0 *(.15)*

BL = baseline; 8 weeks = after 8 weeks of intervention; %Change = percentage change from baseline to post-intervention ((post-intervention − baseline) × 100 / baseline); pre (baseline) vs. post (after 8 weeks of intervention); d = Cohen's d; BC = body composition; FFM = fat-free mass; * $p < .05$; ** $p < .001$; BW = body weight (kg)

Body Circumference Increases Significantly with Weight Training

Paired-sample analyses showed a significant increase in chest circumference, upper arm circumference (left and right), and thigh circumference (left and right) across all experimental groups: **Chest**: 2 Days/week Training Group; baseline (88.28 ± 4.24) to post-test (91.38 ± 3.70), $t(9) = 5.484$, $p < .001$, $d = .1.73$; 3 Days/week Training Group; baseline (92.79 ± 5.83) to post-test (96.47 ± 8.65), $t(9) = 3.055$, $p < .05$, $d = .97$; 4 Days/week Training Group; baseline (89.79 ± 4.01) to post-test (93.30 ± 3.97), $t(10) = 8.940$, $p < .001$, $d = .2.69$. **Upper left arm**: 2 Days/week Training Group; (28.19 ± 2.20) to post-test (29.01 ± 2.00), $t(9) = 2.768$, $p < .05$, $d = .88$; 3 Days/week Training Group; (29.64 ± 2.54) to post-test (30.73 ± 2.66), $t(9) = 6.635$, $p < .001$, $d = 2.10$; 4 Days/week Training Group; (29.56 ± 3.03) to post-test (30.17 ± 3.02), $t(10) = 2.726$, $p < .05$, $d = .82$. **Upper right arm**: 2 Days/week Training Group; baseline (28.56 ± 2.15) to post-test (29.20 ± 1.88), $t(9) = 2.384$, $p < .05$, $d = .75$; 3 Days/week Training Group; 29.72 ± 2.60) to post-test (30.77 ± 2.47), $t(9) = 7.584$, $p < .001$, $d = 2.40$; 4 Days/week Training Group; baseline (29.91 ± 3.13) to post-test (30.55 ± 2.84), $t(10) = 5.513$, $p < .001$, $d = 1.66$. **Left thigh**: 2 Days/week Training Group; baseline (53.56 ± 3.34) to post-test (53.81 ± 3.37), $t(9) = 4.792$, $p < .05$, $d = 1.52$; 3 Days/week Training Group; baseline (54.90 ± 2.92) to post-test (55.34 ± 2.88), $t(9) = 7.833$, $p < .05$, $d = 2.48$; 4 Days/week Training Group; baseline (54.52 ± 4.57) to post-test (54.90 ± 4.54), $t(10) = 9.536$, $p < .001$, $d = 2.88$. **Right thigh**: 2 Days/week Training Group; baseline (52.92 ± 2.48) to post-test (53.33 ± 2.57), $t(9) = 7.499$, $p < .001$, $d = 2.37$; 3 Days/week Training Group; baseline (55.60 ± 3.05) to post-test (55.98 ± 3.15), $t(9) = 6.626$, $p < .001$, $d = 2.09$; 4 Days/week Training Group; baseline (55.05 ± 4.09) to post-test (55.46 ± 4.06), $t(10) = 8.057$, $p < .001$, $d = 2.43$ (Table 2). No significant changes were observed in the control group.

ANCOVA was used to compare training effects among the experimental groups, adjusting for pre-training body circumference values. The results showed that all weight training groups exhibited significantly greater increases in body measurements compared to the control group, with the exception of the Upper Left Arm circumference in the 2 and 4 days/week training groups, which did not differ significantly from the control group. Furthermore, no significant differences were observed among the three exercise groups. The ANCOVA results were as follows. **Chest circumference**: $F(3, 36) = 4.121$, $p < .05$, $\eta^2p = .256$. **Upper left arm circumference**: $F(3, 36) = 4.430$, $p < .05$, $\eta^2p = .270$. **Upper right arm circumference**: $F(3, 36) = 10.406$, $p < .001$, $\eta^2p = .464$. **Left thigh circumference**: $F(3, 36) = 14.861$, $p < .001$, $\eta^2p = .553$. **Right thigh circumference**: $F(3, 36) = 16.246$, $p < .001$, $\eta^2p = .575$. Detailed results are presented in Table 3.

Weight Training Directly Improves Muscle Strength

All experimental groups showed statistically significant increases in 1RM barbell chest press (kg), 1RM barbell chest press / BW, 1RM barbell row (kg), 1RM barbell row / BW, 1RM Smith machine squat (kg), 1RM Smith machine squat / BW, hip thrust (kg), and hip thrust / BW. **1RM Barbell Chest Press (kg)**: *2 Days/week Training Group*; Baseline (48.97 ± 11.23) to post-test (53.26 ± 15), $t(9) = 2.973$, $p < .05$, $d = .94$); *3 Days/week Training Group*; Baseline (51.69 ± 10.43) to post-test (56.88 ± 11.87), $t(9) = 7.643$, $p < .001$, $d = 2.42$; *4 Days/week Training Group*: Baseline (50.94 ± 6.18) to post-test (54.89 ± 6.18), $t(10) = 8.008$, $p < .001$, $d = 2.42$). **1RM Barbell Chest Press / BW**: *2 Days/week Training Group*: Baseline (0.74 ± 0.13) to post-test (0.79 ± 0.19), $t(9) = 2.567$, $p < .05$, $d = .81$; 3 *Days/week Training Group*: Baseline (0.78 ± 0.13) to post-test (0.84 ± 0.14), $t(9) = 9.279$, $p < .001$, $d = 2.93$; *4 Days/week Training Group*: Baseline (0.78 ± 0.08) to post-test (0.83 ± 0.08), $t(10) = 6.212$, $p < .001$, $d = 1.87$. **1RM Barbell Row (kg)**: *2 Days/week Training Group*: Baseline (44.25 ± 6.48) to post-test (50.80 ± 6.98), $t(9) = 6.774$, $p < .001$, $d = 2.14$); *3 Days/week Training Group*: Baseline (46.74 ± 8.49) to post-test (54.13 ± 9.55), $t(9) = 6.895$, $p < .001$, $d = 2.18$; *4 Days/week Training Group*: Baseline (45.37 ± 9.95) to post-test (52.33 ± 11.21), $t(10) = 9.209$, $p < .001$, $d = 2.77$. **1RMBarbell Row / BW**: *2 Days/week Training Group*: Baseline (0.67 ± 0.07) to post-test (0.76 ± 0.19), $t(9) = 6.068$, $p < .001$, $d = 1.92$; 3 *Days/week Training Group*: Baseline (0.71 ± 0.13) to post-test (0.81 ± 0.72), $t(9) = 6.016$, $p < .001$, $d = 1.90$; 4 *Days/week Training Group*: Baseline (0.69 ± 0.13) to post-test (0.80 ± 0.141), $t(10) = 7.611$, $p < .001$, $d = 2.29$. **1RMSmith Machine Squat (kg)**: *2 Days/week Training Group*: Baseline (61.78 ± 15.41) to post-test (81.17 ± 13.88), $t(9) = 4.507$, $p < .05$, $d = 1.43$; *3 Days/week Training Group*: Baseline (63.86 ± 13.46) to post-test (83.75 ± 10.11), $t(9) = 5.481$, $p < .001$, $d = 1.73$; *4 Days/week Training Group*: Baseline (53.74 ± 10.00) to post-test (74.38 ± 13.08), $t(10) = 10.139$, $p < .001$, $d = 3.06$. **1RMSmith Machine Squat / BW**: *2 Days/week Training Group*: Baseline (0.93 ± 0.18) to post-test (1.23 ± 0.23), $t(9) = 4.167$, $p < .05$, $d = 1.32$; *3 Days/week Training Group*: Baseline (0.95 ± 0.11) to post-test (1.26 ± 0.17), $t(9) = 4.494$, $p < .05$, $d = 1.42$; *4 Days/week Training Group*: Baseline (0.83 ± 0.18) to post-test (1.13 ± 0.24), $t(10) = 9.682$, $p < .001$, $d = 2.92$. **1RM Hip Thrust (kg)**: *2 Days/week Training Group*: Baseline (61.21 ± 6.77) to post-test (68.93 ± 6.76), $t(9) = 5.136$, $p < .05$, $d = 1.62$; *3 Days/week Training Group*: Baseline (60.14 ± 8.60) to post-test (68.38 ± 8.30), $t(9) = 6.676$, $p < .001$, $d = 2.11$; *4 Days/week Training Group*: Baseline (60.07 ± 8.56) to post-test (68.50 ± 6.30), $t(10) = 6.520$, $p < .001$, $d = 2.11$. Hip Thrust / BW: *2 Days/week Training Group*: Baseline (0.93 ± 0.10) to post-test (1.04 ± 0.09), $t(9) = 2.567$, $p < .05$, $d = 1.43$; *3 Days/week Training Group*: Baseline (0.91 ± 0.10) to post-test (1.02 ± 0.09), $t(9) = 9.279$, $p < .001$, $d = 1.68$; *4 Days/week Training Group*: Baseline (0.92 ± 0.11) to post-test (1.04 ± 0.12), $t(10) = 6.212$, $p < .05$, $d = 1.47$ (Table 2). No significant changes in any 1RM measures were observed in the control group.

ANCOVA, adjusted for baseline 1RM, indicated a statistically significant difference between at least one group for all muscle strength variables. **Barbell Chest Press**: $F(3,36) = 6.030$, $p < .05$, $\eta^2 p = 0.302$. **Barbell Row**: $F(3,36) = 16.771$, $p < .001$, $\eta^2 p = .583$. **Smith Machine Squat**: $F\ (3,36) = 12.817$, $p < .001$, $\eta^2 p = .516$. **Hip Thrust**: $F(3,36) = 13.227$, $p < .001$, $\eta^2 p = .524$. Post-hoc analysis with Bonferroni correction showed that all three RT frequencies (2, 3, and 4 days/week) produced significant improvements in 1RM muscle strength across all exercises when compared to the non-training control group. However, no statistically significant differences were detected among the three training groups.

Table3. Intergroup comparisons of body composition, body circumferences, and muscle strength (1 RM)

Variable	covariate	2 days/week	3 days/week	4 days/week	Control	ANCOVA	
		Mean (95% CI)	Mean (95% CI)	Mean (95% CI)	Mean (95% CI)	*F*	η^2_p
BC							
Weight (kg)	65.92	66.70 (66.1; 67.31)	66.68 (66.07; 67.29)	66.64 (66.07; 67.22)	66.25 (65.64; 66.86)	.517	.041
FFM (kg)	53.06	53.83 (53.41; 54.25) (4)	53.80 (53.38; 54.23) (4)	53.79 (53.39; 54.20) (4)	53.09 (52.66; 53.51) (1)(2)(3)	2.998*	.200
% Fat	14.74	14.62 (13.91; 15.34)	14.76 (14.09; 15.48)	14.73 (14.05; 15.41)	15.03 (14.31; 15.74)	.233	.019
Circumference (cm)							
Chest	90.77	94.02 (92.52; 95.51) (4)	94.23 (92.84; 95.80) (4)	94.33 (92.95; 95.72) (4)	91.32 (89.86; 92.79) (1)(2)(3)	4.121*	.256
Upper left arm	29.10	29.89 (29.46; 30.12)	30.21 (29.78; 30.63) (4)	29.73 (29.33; 30.14)	29.15 (28.73; 29.58) (2)	4.430*	.270
Upper right arm	29.54	29.92 (29.61; 30.23) (4)	30.42 (30.11; 30.73) (4)	30.02 (29.72; 30.31) (4)	29.23 (28.92; 29.54) (1)(2)(3)	10.41**	.464
Left thigh	54.36	54.61 (54.51; 54.71) (4)	54.80 (54.70; 54.90) (4)	54.74 (54.64; 54.84) (4)	54.37 (54.27; 54.47) (1)(2)(3)	14.86**	.553
Right thigh	54.68	55.11 (55; 55.22) (4)	55.05 (54.95; 55.16) (4)	55.10 (54; 55.20) (4)	54.67 (54.56; 54.78) (1)(2)(3)	16.25**	.575
Muscle strength 1RM							
Barbell chest press	49.90	54.33 (52.27; .56.38) (4)	54.83 (52.77; 56.90) (4)	53.70 (51.74; 55.66) (4)	49.72 (47.66; 51.79) (1)(2)(3)	6.030*	.302
Barbell row	46.25	52.87 (51.15; 54.59) (4)	53.62 (51.92; 55.33) (4)	53.45 (51.81; .55.08) (4)	46.39 (44.67; 48.12) (1)(2)(3)	16.771**	.583
Smith machine squat	59.15	79.41 (73.82; 85.00) (4)	80.60 (74.94; 86.26) (4)	78.00 (72.57; 83.45) (4)	59.80 (54.23; .65.36) (1)(2)(3)	12.817**	.516
Hip thrust	60.50	68.33 (66.07; 70.60) (4)	68.45 (66.20; .70.71) (4)	68.86 (66.70; 71.01) (4)	60.49 (58.23; 62.76) (1)(2)(3)	13.227**	.524

Covariate = ANCOVA adjustment from baseline; * = p < .05; ** = p < .001; BC = body composition; FFM = fat-free mass; (1) = significant with 2 days/week; (2) = significant with 3 days/week; (3) = significant with 4 days/week; (4) = significant with control group; BW = body weight (kg).

Discussion

This study used an Equal-Set and Intensity-Matched design to investigate how different RT frequencies (2, 3, and 4 days/week) influence outcomes in previously untrained young males. The findings indicate that training frequency did not produce distinct differences in improvements in muscular strength, body circumference, or FFM, provided that the total number of sets and relative intensity were held constant. Nonetheless, all RT groups demonstrated significant gains across these measures compared to the non-training control group. These results underscore the importance of RT for improving physical fitness, particularly muscular strength and body composition (ACSM, 2018; Schoenfeld, 2010; Hong & Kim, 2018).

Effect of Resistance Training on Muscle Strength

The findings demonstrated that RT, irrespective of frequency (2, 3, or 4 days/week), produced statistically significant increases in muscular strength, as assessed by 1RM in the barbell chest press, barbell row, Smith machine squat, and hip thrust, compared with the control group. These results confirm that RT is an effective strategy for enhancing muscular strength and are consistent with extensive evidence highlighting the role of resistance exercise in promoting neuromuscular adaptations (Schoenfeld, 2010; Suchomel et al., 2018). In untrained or novice individuals, initial strength gains are largely driven by neural adaptations, such as increased motor unit recruitment, enhanced rate coding, and improved synchronization of firing (Enoka, 2008; Kessinger et al., 2020). With continued RT, muscular hypertrophy becomes a more prominent contributor to strength development (Damas et al., 2015; Gomes et al., 2019). These findings highlight the substantial impact of weight training on muscular strength, aligning with the work of Grgic et al. (2018), who reported an effect size greater than 0.8 for strength improvements when training occurred at least twice per week.

The results also showed no statistically significant differences in muscular strength among the RT groups with different frequencies (2, 3, or 4 days/week) when the number of sets and relative intensity were matched. This finding aligns with previous volume-matched research and reinforces the idea that the overall training stimulus—rather than the specific frequency of distribution—is the key determinant of strength development ([2]Schoenfeld et al., 2019; Lasevicius et al., 2018). The underlying physiological mechanism is that performing an adequate number of sets at sufficient intensity effectively stimulates muscle protein synthesis (MPS) signaling pathways, regardless of whether the training stimulus is distributed frequently with a lower per-session set volume or less frequently with a higher per-session set volume. As long as the total weekly sets at an appropriate intensity are sufficient, similar neuromuscular and structural muscle adaptations can be achieved (Schoenfeld, 2010; Wackerhage et al., 2019)

This indicates that for untrained young adults, the total training stimulus—determined by matched sets and intensity—has a greater impact than the way it is distributed across the week.

Effects of Resistance Training on Muscle Hypertrophy: Anthropometric Parameters (Body Circumference)

The findings demonstrated that all weight training groups experienced statistically significant increases in body circumferences, including the chest, upper arms, and thighs, from pre- to post-training. Each of the three training frequencies produced significant improvements in these measurements compared with the control group. These increases reflect the beneficial effects of RT on muscle hypertrophy, a key adaptation to mechanical tension and metabolic stress (ACSM, 2018; Schoenfeld, 2010).

Muscle hypertrophy primarily results from increased MPS, which promotes the growth of myofibrils and interstitial connective tissue in muscle fibers, ultimately increasing muscle cross-sectional area (CSA) (Schoenfeld, 2010; Wackerhage et al., 2019). In this study, increases in muscle size, assessed via body circumferences, ranged from 0.46% to 3.97%. Notably, no significant differences were observed in body circumference changes among groups with different training frequencies (2–4 times per week), indicating that frequency had little impact when total sets and relative intensity were matched. These findings suggest that when total sets and training intensity are consistent, the number of weekly training sessions has minimal impact on muscle size or body circumference (Schoenfeld et al., 2019; Grgic et al., 2021; Johnsen et al., 2021). This underscores the importance of performing an adequate volume of sets at sufficient intensity to stimulate hypertrophic adaptations. Therefore, training frequency is primarily a way to distribute the stimulus rather than change its overall effect; maintaining consistent intensity produces similar growth responses (Schoenfeld, 2010; Wackerhage et al., 2019). For untrained individuals, RT 2–3 days/week is sufficient to increase muscle CSA when the program is consistent and properly structured. In contrast, individuals with prior RT experience may require 4–6 sessions per week to achieve comparable gains (Kraemer & Ratamess, 2004; Schoenfeld et al., 2016). It is also important to note that the early increases in muscle size observed in untrained young men during the first 1–3 weeks of training may not reflect true hypertrophy; these initial changes could primarily result from water retention owing to muscle damage, with actual muscle growth occurring later (Damas et al., 2016). Overall, this study confirmed that all RT groups exhibited significant increases in body circumference across multiple sites compared with the control group, reinforcing the effectiveness of weight training for promoting muscle hypertrophy.

Effect of Resistance Training on Body Composition

This study found that RT led to significant increases in BW and FFM across all training groups, while the %Fat remained unchanged. Post-hoc analysis confirmed that all RT groups (2, 3, and 4 days/week) demonstrated significantly greater gains in FFM compared with the control group, highlighting the effectiveness of RT in promoting the development of fat-free tissue. The observed increases in FFM are primarily attributed to gains in muscle mass, which are critical for overall health and performance because greater muscle mass increases resting metabolic rate and improves body composition (Hong & Kim, 2018). These physiological adaptations in previously untrained individuals further emphasize the effectiveness of RT interventions (Mao et al., 2023; Suchomel et al., 2018).

However, no statistically significant differences in FFM were observed among the RT groups (2, 3, and 4 days/week). This finding is consistent with previous volume-equated studies, which suggest that varying training frequency does not substantially influence muscle growth or FFM when the total number of sets at matched intensity is equivalent (Schoenfeld et al., 2019; Grgic et al., 2018; Johnsen et al., 2021). These results indicate that the total weekly training volume performed at an appropriate intensity is likely a more critical factor for stimulating muscle growth than frequency, which aligns with our observations for strength and body circumference. Regarding BW, significant within-group increases were observed across all training groups, indicating that the gains in BW were primarily attributed to increased muscle mass rather than changes in fat mass. Similarly, no statistically significant changes in %Fat were observed in either the experimental or control groups. This finding aligns with previous research (Mangine et al., 2015), suggesting that RT alone may not be sufficient to elicit significant reductions in body fat because fat loss generally requires a negative energy balance achieved through dietary modifications or additional cardiovascular exercise. Since resistance training focuses on developing muscle size and strength, while aerobic or cardio exercise emphasizes developing the circulatory system and burning body fat (Rugbumrung et al., 2025)

While our study's findings showed significant increases in both Fat-Free Mass (FFM) and muscular strength (1RM) across all three RT groups, aligning with the idea that FFM is positively correlated with muscle strength (Acharya et al., 2022), other research has reported no significant link between FFM and muscle strength (Bagheri et al., 2024; Moore et al., 2020). Therefore, the data from this study cannot definitively establish or interpret that the increase in FFM is directly related to the rise in muscular strength among the participants. However, the main focus of this study was training frequency, and the key finding is that all RT groups experienced similar improvements in strength and body composition. This suggests that training frequency did not significantly influence muscle growth or strength when total volume was controlled. Consequently, the optimal frequency depends on personal preference, and novice trainees can choose their weekly training frequency per muscle group based on their own convenience ([2]Schoenfeld et al., 2019).

Conclusion

This RCT confirms that RT is highly effective for improving physical fitness in previously untrained young adults. The study demonstrated that training at any frequency 2, 3, or 4 days/week produced significant physiological improvements compared with the control group, including increases in muscular strength, changes in body measurements, and gains in FFM. Notably, the Equal-Set and Intensity-Matched design revealed that variations in training frequency (2, 3, or 4 days/week) did not result in statistically significant differences in strength, muscle size, or body composition outcomes. These findings strongly support the notion that an adequate overall training stimulus—defined by the total number of sets performed at an appropriate relative intensity is the primary driver of physiological adaptations. The practical implication is that beginners and individuals with limited time can select a training frequency that fits their schedule without compromising strength or muscle gains, which may improve exercise adherence. This insight simplifies workout planning and facilitates the design of effective, accessible RT programs aimed at maximizing health-related physical fitness, particularly for beginners. Future research should examine the effects of RT over longer durations, such as 12–24 weeks, to investigate potential differences as participants progress to an intermediate training level. Additionally, applying the Equal-Set and Intensity-Matched principles to other populations—including untrained women, obese individuals, older adults, or clinical populations—is recommended to broaden the generalizability and confirm the time-efficient effectiveness of this training approach.

Limitations of the Study

This study has several limitations that should be considered when interpreting the results. First, the intervention lasted only 8 weeks. Although untrained individuals can experience rapid early gains in strength and muscle size, the long-term effects of varying training frequency—when total set volume and relative intensity are matched—may differ, and more sustained physiological adaptations may require a longer intervention period. Second, the study involved only untrained young adults, which limits the generalizability of the findings to other populations, such as older adults, highly trained athletes, or individuals with pre-existing medical conditions, who may respond differently to RT. Third, although total set number and relative intensity were carefully controlled, the distribution of training volume—based on specific RM ranges—varied among the frequency groups. This resulted in slight differences in the total repetitions performed by each group, which could have influenced the results. Additionally, the study did not assess cumulative fatigue or recovery between sessions. Fourth, body circumference was obtained using anthropometric techniques, and body composition (FFM, %Fat) was assessed using BIA. These methods may be less sensitive to detecting small changes compared with more advanced techniques, such as DEXA or ultrasound, particularly over a relatively short intervention period. Finally, although participants were encouraged to maintain their usual diet and lifestyle, their dietary intake was not strictly monitored or recorded, which could have influenced the physiological adaptations to the training program.

Suggestions for Future Research

Several recommendations emerge for future research based on the current study's findings and limitations. First, longer-duration interventions, such as those lasting six months or more, would provide insight into the long-term effects of different training frequencies while maintaining matched total sets and relative intensity, enhancing our understanding of sustainable strength and muscle growth. Second, future studies should include more diverse populations—including older adults, experienced athletes, or individuals with specific training goals—to improve the generalizability of the findings and ensure broader applicability across different demographic groups. Third, future research should evaluate qualitative factors such as accumulated fatigue, muscle recovery time, sleep quality, and ratings of perceived exertion, because these variables may be influenced by training frequency even when total sets and relative intensity are constant. Fourth, using more accurate assessment methods, such as DEXA or BIA, is recommended to obtain accurate and detailed body composition measurements, which would improve understanding of how different training protocols affect physiological adaptations.

Conflicts of interest The authors declare no conflicts of interest.

Acknowledgement This study was supported by a grant from the School of Science, University of Phayao, Thailand. The authors would like to thank Falcon Scientific Editing (https://falconediting.com) for proofreading the English language in this paper

Declaration of generative AI and AI-assisted technologies in the writing process

The authors used Google Gemini and Grammarly to refine the writing, improve grammar, and enhance clarity in this manuscript. These tools were not used to generate content, images, or tables. All material was reviewed and edited by the authors, who take full responsibility for the final publication.

References:

American College of Sports Medicine. (2018). *ACSM's guidelines for exercise testing and prescription* (10th ed.). Wolters Kluwer.

Acharya, S., Patnaik, M., Kumar, K. P., & Mishra, S. P. (2022). Correlation of body fat percentage, fat mass, fat free mass, and fat free mass index with hand grip strength in healthy young males in a tertiary health care center of South Odisha. *National Journal of Physiology, Pharmacy and Pharmacology*, *12*(2), 116–120.

Arazi, H., Asadi, A., Gentil, P., Ramírez-Campillo, R., Jahangiri, P., Ghorbani, A., Hackney, A. C., & Zouhal, H. (2021). Effects of different resistance training frequencies on body composition and muscular performance adaptations in men. *PeerJ*, *21*(9), e10537. https://doi.org/10.7717/peerj.10537

Arazi, H., Khoshnoud, A., Asadi, A., & Tufano, J. J. (2021). The effect of resistance training set configuration on strength and muscular performance adaptations in male powerlifters. *Scientific Reports*, *11*(1), 7844.

Bagheri, R., Karimi, Z., Camera, D. M., Scott, D., Bashirzad, M. Z., Sadeghi, R., Kargarfard, M., & Dutheil, F. (2024). Association between changes in lean mass, muscle strength, endurance, and power following resistance or concurrent training with differing high protein diets in resistance-trained young males. *Frontiers in Nutrition*, *11*, Article 1439037. https://doi.org/10.3389/fnut.2024.1439037

Bauman, A. E., Reis, R. S., Sallis, J. F., Wells, J. C., Loos, R. J. F., Martin, B. W., & Lancet Physical Activity Series Working Group. (2012). Correlates of physical activity: Why are some people physically active and others not? *The Lancet*, *380*(9838), 258–271. https://doi.org/10.1016/S0140-6736(12)60735-1

Chaves, T. S., Pires de Campos Biazon, T. M., Marcelino Eder Dos Santos, L., & Libardi, C. A. (2020). Effects of resistance training with controlled versus self-selected repetition duration on muscle mass and strength in untrained men. PeerJ, 8, e8697. https://doi.org/10.7717/peerj.8697

Cigerci, A. E., & Genc, H. (2020). The effect of strength training with different frequency on untrained university students. *Physical Education of Students*, *24*(4), 186–193. https://doi.org/10.15561/20755279.2020.0401

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences*. Lawrence Erlbaum Associates Inc.

Cuthbert, M., Haff, G. G., Arent, S. M., Ripley, N., McMahon, J. J., Evans, M., & Comfort, P. (2021). Effects of variations in resistance training frequency on strength development in well-trained populations and implications for in-season athlete training: A systematic review and meta-analysis. *Sports Medicine*, *51*(9), 1967–1982. https://doi.org/10.1007/s40279-021-01460-7

Damas, F., Phillips, S. M., Lixandrão, M. E., Vechin, F. C., Libardi, C. A., Roschel, H., Tricoli, V., & Ugrinowitsch, C. (2016). Early resistance training-induced increases in muscle cross-sectional area are concomitant with edema-induced muscle swelling. *European Journal of Applied Physiology*, *116*(1), 49–56. https://doi.org/10.1007/s00421-015-3243-4

Damas, F., Phillips, S., Vechin, F. C., & Ugrinowitsch, C. (2015). A review of resistance training-induced changes in skeletal muscle protein synthesis and their contribution to hypertrophy. Sports Medicine, 45(6), 801–807. https://doi.org/10.1007/s40279-015-0320-0

Enoka, R. M. (2008). *Neuromechanics of human movement* (4th ed.). Human Kinetics.

Gomes, G. K., Franco, C. M., Nunes, P. R. P., & Orsatti, F. L. (2019). High-frequency resistance training is not more effective than low-frequency resistance training in increasing muscle mass and strength in well-trained men. *Journal of Strength and Conditioning Research*, *33*(Suppl 1), S130–S139. https://doi.org/10.1519/JSC.0000000000002559

Grgic, J., Lazinica, B., Schoenfeld, B. J., & Pedisic, Z. (2020). Test-retest reliability of the one-repetition maximum (1RM) strength assessment: A systematic review. *Sports Medicine-Open*, *6*(1), 31. https://doi.org/10.1186/s40798-020-00260-z

Grgic, J., Schoenfeld, B. J., Davies, T. B., Lazinica, B., Krieger, J. W., & Pedisic, Z. (2018). Effect of resistance training frequency on gains in muscular strength: A systematic review and meta-analysis. *Sports*

Medicine, *48*(5), 1207–1220. https://doi.org/10.1007/s40279-018-0872-x

Hamarsland, H., Moen, H., Skaar, O. J., Rønnestad, B. R., & Raastad, T. (2022). Equal-volume strength training with different training frequencies induces similar muscle hypertrophy and strength improvement in trained participants. *Frontiers in Physiology*, *12*, 789403. https://doi.org/10.3389/fphys.2021.789403

Hansen, R. K., Jørgensen, R. R., Nielsen, P. S., Osuna-Florentz, P., & Kristiansen, M. (2017). Effects of resistance training frequency on short-term neuromuscular adaptations in untrained participants. *Advances in Physical Education*, *7*(4), 383–398. https://doi.org/10.4236/ape.2017.74032

Hong, A. R., & Kim, S. W. (2018). Effects of resistance exercise on bone health. *Endocrinology and Metabolism*, *33*(4), 435–444. doi: 10.3803/EnM.2018.33.4.435.

Hughes, D. C., Ellefsen, S., & Baar, K. (2018). Adaptations to endurance and strength training. *Cold Spring Harbor Perspectives in Medicine*, *8*(6), a029769. https://doi.org/10.1101/cshperspect.a029769

Johnsen, E., & van den Tillaar, R. (2021). Effects of training frequency on muscular strength for trained men under volume matched conditions. *PeerJ*, *9*, e10781. https://doi.org/10.7717/peerj.10781

Kessinger, T. K., Melton, B., Miyashita, T., & Ryan, G. (2020). The effectiveness of frequency-based resistance training protocols on muscular performance and hypertrophy in trained males: A critically appraised topic. *Journal of Sport Rehabilitation*, *29*(7), 1024–1031. https://doi.org/10.1123/jsr.2019-0491

Kraemer, W. J., & Ratamess, N. A. (2004). Fundamentals of resistance training: Progression and exercise prescription. *Medicine & Science in Sports & Exercise*, *36*(4), 674–688.

Lasevicius, T., Schoenfeld, B. J., Silva-Batista, C., Barros, T. de S., Aihara, A. Y., Brendon, H., Longo, A. R., Tricoli, V., Peres, B. de A., & Teixeira, E. L. (2022). Muscle failure promotes greater muscle hypertrophy in low-load but not in high-load resistance training. Journal of Strength and Conditioning Research, 36(2), 346–351. https://doi.org/10.1519/JSC.0000000000003454

Lasevicius, T., Ugrinowitsch, C., Schoenfeld, B. J., Roschel, H., Tavares, L. D., De Souza, E. O., Laurentino, G., & Tricoli, V. (2018). Effects of different intensities of resistance training with equated volume load on muscle strength and hypertrophy. *European Journal of Sport Science*, *18*(6), 772–780. https://doi.org/10.1080/17461391.2018.1450898

Mangine, G. T., Hoffman, J. R., Gonzalez, A. M., Townsend, J. R., Wells, A. J., Jajtner, A. R., Fragala, M. S., Miramonti, A. A., & Stout, J. R. (2015). The effects of training volume and intensity on improvements in muscular strength and body composition in trained men. *Physiological Reports*, *3*(8), e12472. https://doi.org/10.14814/phy2.12472

Mao, J., Wang, T., Wang, Y., & Kuan, G. (2023). Effect of resistance training programs differing in set structure on muscular hypertrophy and performance in untrained young men. *Frontiers in Physiology*, *14*, 1301535. https://doi.org/10.3389/fphys.2023.1301535

Mcleod, J. C., Currier, B. S., Lowisz, C. V., & Phillips, S. M. (2024). The influence of resistance exercise training prescription variables on skeletal muscle mass, strength, and physical function in healthy adults: An umbrella review. *Journal of Sport and Health Science*, *13*(1), 47–60. https://doi.org/10.1016/j.jshs.2023.06.005

Moore, B. A., Bemben, D. A., Lein, D. H., Bemben, M. G., & Singh, H. (2020). Fat mass is negatively associated with muscle strength and jump test performance. *The Journal of Frailty & Aging*, *9*(4), 214–218. https://doi.org/10.14283/jfa.2020.11

Moesgaard, L., Beck, M. M., Christiansen, L., Aagaard, P., & Lundbye-Jensen, J. (2022). Effects of periodization on strength and muscle hypertrophy in volume-equated resistance training programs: A systematic review and meta-analysis. *Sports Medicine*, *52*(7), 1647–1666. https://doi.org/10.1007/s40279-021-01636-1

Neves, R. P., Vechin, F. C., Teixeira, E. L., da Silva, D. D., Ugrinowitsch, C., Roschel, H., de Siqueira, K. R., & Lasevicius, T. (2022). Effect of different training frequencies on maximal strength performance and muscle hypertrophy in trained individuals—a within-subject design. *PLoS ONE*, *17*(10), e0276154. https://doi.org/10.1371/journal.pone.0276154

Paluch, A. E., Boyer, W. R., Franklin, B. A., Laddu, D., Lobelo, F., Lee, D. C., & McDermott, M. M. (2024). Resistance exercise training in individuals with and without cardiovascular disease: 2023 update: A scientific statement from the American Heart Association. *Circulation*, *149*(3), e217–e231. https://doi.org/10.1161/CIR.0000000000001189

Rhea, M. R., Alvar, B. A., Ball, S. D., & Burkett, L. N. (2002). A comparison of linear and daily undulating periodized programs with equated volume and intensity for strength. *Journal of Strength and Conditioning Research*, *16*(2), 250–255.

Rugbumrung, M., Rukbumrung, T., Impanya, S., & Thurayot, A. (2025). Effects of aerobic exercise on physical fitness in obesity using Fox vs. Tanaka's maximum heart rate and percentage vs. Karvonen methods. *International Journal of Exercise Science*, *18*(8), 695–711. https://doi.org/10.70252/XEPS8890

Schoenfeld, B. J. (2010). The mechanisms of muscle hypertrophy and their application to resistance training. *Journal of Strength and Conditioning Research*, *24*(10), 2857–2872. doi: 10.1519/JSC.0b013e3181e840f3.

[1]Schoenfeld, B. J., Contreras, B., Krieger, J., Grgic, J., Haun, C., Franzo, T., & Willardson, J. M. (2019). Resistance training volume enhances muscle hypertrophy but not strength in trained men. *Medicine & Science in Sports & Exercise*, *51*(1), 94–103. doi: 10.1249/MSS.0000000000001764.

[2]Schoenfeld, B. J., Grgic, J., & Krieger, J. W. (2019). How many times per week should a muscle be trained to maximize muscle hypertrophy? A systematic review and meta-analysis of studies examining the effects of resistance training frequency. *Journal of Sports Sciences*, *37*(11), 1286-1295. doi: 10.1080/02640414.2018.1555906.

Schoenfeld, B. J., Ogborn, D. I., & Krieger, J. W. (2016). Effects of resistance training frequency on measures of muscle hypertrophy: A systematic review and meta-analysis. *Sports Medicine*, *46*(11), 1689–1697. doi: 10.1007/s40279-016-0543-8.

Staniszewski, M., Mastalerz, A., & Urbanik, C. (2020). Effect of a strength or hypertrophy training protocol, each performed using two different modes of resistance, on biomechanical, biochemical and anthropometric parameters. *Biology of Sport*, *37*(1), 85–91. doi: 10.5114/biolsport.2020.92517.

Suchomel, T. J., Nimphius, S., & Stone, M. H. (2018). The importance of muscular strength: Training considerations. *Sports Medicine*, *48*(4), 765–785. https://doi.org/10.1007/s40279-018-0862-z

Tan, J., Krasilshchikov, O., Kuan, G., Hashim, H. A., Aldhahi, M. I., Al-Mhanna, S. B., & Badicu, G. (2023). The effects of combining aerobic and heavy resistance training on body composition, muscle hypertrophy, and exercise satisfaction in physically active adults. *Healthcare (Basel)*, *11*(17), 2443. https://doi.org/10.3390/healthcare11172443

van den Tillaar, R., & Ball, N. (2020). Push-ups are able to predict the bench press 1-RM and constitute an alternative for measuring maximum upper body strength based on load-velocity relationships. *Journal of Human Kinetics*, *73*, 7–18. doi: 10.2478/hukin-2019-0133

Wackerhage, H., Schoenfeld, B. J., Hamilton, D. L., Lehti, M., & Hulmi, J. J. (2019). Stimuli and sensors that initiate skeletal muscle hypertrophy. *Journal of Applied Physiology*, *126*(1), 30–43. https://doi.org/10.1152/japplphysiol.00685.2018

Wang, J., Thornton, J. C., Kolesnik, S., & Pierson, R. N. (2000). Anthropometry in body composition: An overview. *Annals of the New York Academy of Sciences*, *904*(1), 317–326. doi: 10.1111/j.1749-6632.2000.tb06474.x.

Wernbom, M., Augustsson, J., & Thomeé, R. (2007). The influence of frequency, intensity, volume and mode of strength training on whole muscle cross-sectional area in humans. *Sports Medicine*, *37*(3), 225–264. https://doi.org/10.2165/00007256-200737030-00004

Yang, S. W., Kim, T. H., & Choi, H. M. (2018). The reproducibility and validity verification for body composition measuring devices using bioelectrical impedance analysis in Korean adults. *Journal of Exercise Rehabilitation*, *14*(4), 621–627. https://doi.org/10.12965/jer.1836284.142

Zaroni, R. S., Brigatto, F. A., Schoenfeld, B. J., Braz, T. V., Benvenutti, J. C., Germano, M. D., & Lima, M. B. (2019). High resistance-training frequency enhances muscle thickness in resistance-trained men. *Journal of Strength and Conditioning Research*, *33*(Suppl 1), S140–S151. doi: 10.1519/JSC.0000000000002643.